

What Is Data Science Life Cycle?

Data Science is the amalgamation of two fields – Data and Science.

Data is any real or imaginary thing and science is nothing but systematic study of world both physical and natural.

So Data Science is nothing but systematic study of data and derivation of knowledge using testable methods to do predictions about the Universe.

In simple words its applying science on data which may be of any size and from any source.

Data has become a new oil that is driving businesses today. That's why understanding the data science project life cycle is crucial.

As a Data Scientist or Machine Learning Engineer or as a Project Manager you must be aware of the important steps.

Data analytics is the process of analyzing raw data in order to draw out meaningful, actionable insights, which are then used to inform and drive smart business decisions.

A data analyst will extract raw data, organize it, and then analyze it, transforming it from incomprehensible numbers into coherent, intelligible information. Having interpreted the data, the data analyst will then pass on their findings in the form of suggestions or recommendations about what the company's next steps should be.

You can think of data analytics as a form of business intelligence, used to solve specific problems and challenges within an organization. It's all about finding patterns in a dataset which can tell you something useful and relevant about a particular area of the business—how certain customer groups behave, for example, or how employees engage with a particular tool.

What's the difference between data analytics and data science?

Key difference between data scientists and data analysts lies in what they do with the data and the outcomes they achieve. A data analyst will seek to answer specific questions or address particular challenges that have already been identified and are known to the business. To do this, they examine large datasets with the goal of identifying trends and patterns. They then

“visualize” their findings in the form of charts, graphs, and dashboards. These visualizations are shared with key stakeholders and used to make informed, data-driven strategic decisions.

A data scientist, on the other hand, considers what questions the business should or could be asking. They design new processes for data modeling, write algorithms, devise predictive models, and run custom analyses. For example: They might build a machine to leverage a dataset and automate certain actions based on that data—and, with continuous monitoring and testing, and as new patterns and trends emerge, improve and optimize that machine wherever possible.

In short: data analysts tackle and solve discrete questions about data, often on request, revealing insights that can be acted upon by other stakeholders, while data scientists build systems to automate and optimize the overall functioning of the business.

What are the different types of data analysis?

Now we have a working definition of data analytics, let's explore the four main types of data analysis: descriptive, diagnostic, predictive, and prescriptive.

Descriptive analytics

Descriptive analytics is a simple, surface-level type of analysis that looks at **what has happened in the past**. The two main techniques used in descriptive analytics are data aggregation and data mining—so, the data analyst first gathers the data and presents it in a summarized format (that's the aggregation part) and then “mines” the data to discover patterns.

The data is then presented in a way that can be easily understood by a wide audience (not just data experts). It's important to note that descriptive

analytics doesn't try to explain the historical data or establish cause-and-effect relationships; at this stage, it's simply a case of determining and describing the "what". Descriptive analytics draws on **descriptive statistics**

Diagnostic analytics

While descriptive analytics looks at the "what", **diagnostic analytics explores the "why"**. When running diagnostic analytics, data analysts will first seek to identify anomalies within the data—that is, anything that cannot be explained by the data in front of them. For example: If the data shows that there was a sudden drop in sales for the month of March, the data analyst will need to investigate the cause.

To do this, they'll embark on what's known as the discovery phase, identifying any additional data sources that might tell them more about why such anomalies arose. Finally, the data analyst will try to uncover causal relationships—for example, looking at any events that may correlate or correspond with the decrease in sales. At this stage, data analysts may use probability theory, regression analysis, filtering, and time-series data analytics.

Predictive analytics

Just as the name suggests, predictive analytics tries to predict **what is likely to happen in the future**. This is where data analysts start to come up with actionable, data-driven insights that the company can use to inform their next steps. Predictive analytics estimates the likelihood of a future outcome based on historical data and probability theory, and while it can never be completely accurate, it does eliminate much of the guesswork from key business decisions.

Predictive analytics can be used to forecast all sorts of outcomes—from what products will be most popular at a certain time, to how much the company revenue is likely to increase or decrease in a given period. Ultimately, predictive analytics is used to increase the business's chances of “hitting the mark” and taking the most appropriate action.

Prescriptive analytics

Building on predictive analytics, prescriptive analytics **advises on the actions and decisions that should be taken.** In other words, prescriptive analytics shows you how you can take advantage of the outcomes that have been predicted. When conducting prescriptive analysis, data analysts will consider a range of possible scenarios and assess the different actions the company might take.

Prescriptive analytics is one of the more complex types of analysis, and may involve working with algorithms, machine learning, and computational modeling procedures. However, the effective use of prescriptive analytics can have a huge impact on the company's decision-making process and, ultimately, on the bottom line.